DISCRETE MATH · PROPERTIES OF RELATIONS

Relation Property Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Relation Property Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. On A = {1, 2, 3}, is R = {(1, 1), (2, 2), (3, 3), (1, 2)} reflexive? Symmetric? Transitive?
   Reflexive yes; symmetric no ((1, 2) without (2, 1)); transitive yes.
2. How would you see "symmetric" in a directed graph?
   Every arrow a → b has a return arrow b → a.

2 Explore use the tool

1. Load "a ≤ b" on {1, 2, 3, 4}. Which properties hold?
   Reflexive, antisymmetric and transitive (a partial order); not symmetric.
2. Build a relation that is reflexive and symmetric but not transitive. Record it and the counterexample.
   For example the loops plus (1, 2), (2, 1), (2, 3), (3, 2): (1, 2), (2, 3) ∈ R but (1, 3) ∉ R.
3. Make the chain (1, 2), (2, 3), (3, 4) and press Transitive closure. Which pairs were added?
   (1, 3), (2, 4), (1, 4): 3 pairs.
4. On {1, ..., 5}, load "a ≡ b (mod 3)". List the equivalence classes.
   [1] = {1, 4}, [2] = {2, 5}, [3] = {3}.

3 Record use the tool and fill in the table

Properties of relations on {1, 2, 3, 4}

Use "Load a relation" or build each one.

Relation	Reflexive	Symmetric	Antisymmetric	Transitive
a = b	yes	yes	yes	yes
a < b	no	no	yes	yes
a divides b	yes	no	yes	yes
same parity	yes	yes	no	yes
{(1, 2), (2, 1)}	no	yes	no	no

4 Explain what did you discover?

1. Reflexive: a ___loop___ at every vertex. Symmetric: every arrow has a ___return___ arrow.
2. Transitive: every two-step path a → b → c has a ___shortcut (direct arrow)___ a → c.
3. An equivalence relation is reflexive, ___symmetric___ and transitive; its classes form a ___partition___ of A.

4. A student says "symmetric and transitive implies reflexive: from (a, b) and (b, a) we get (a, a)." Find the flaw and give a counterexample.
   The argument needs some b with (a, b) ∈ R. An element related to nothing gets no loop. On {1, 2, 3}, R = {(1, 1), (1, 2), (2, 1), (2, 2)} is symmetric and transitive but (3, 3) ∉ R.

5 Apply a new situation, no tool

1. On the integers, is R = {(m, n) : m − n is even} an equivalence relation? Describe the classes.
   Yes (reflexive: 0 is even; symmetric: n − m = −(m − n); transitive: sum of evens is even). Classes: evens and odds.
2. How many ordered pairs are in the equivalence relation for the partition {{1, 2, 3}, {4, 5}}?
   $3^2 + 2^2 = 13$.
3. Find the transitive closure of {(1, 2), (2, 3), (3, 1)} on {1, 2, 3}.
   All 9 pairs (every element reaches every element, including itself).
4. Is "is a sibling of" transitive? Is it reflexive?
   Not reflexive (no one is their own sibling). Not transitive: a is a sibling of b and b of a, but a is not a sibling of a.

Teacher notes

Intended discovery: Each property has a digraph test and is disproved by one witness; closures add the fewest pairs to force a property; equivalence relations correspond to partitions; symmetric and antisymmetric can hold together.

Likely misconceptions:

- Thinking symmetric and antisymmetric are opposites.
- Missing chains that return to the start (a → b → a needs a loop for transitivity).
- Believing symmetric + transitive forces reflexive.

Answers that can vary: Explore 2 relations vary; any correct counterexample is acceptable.